

PPO, GRPO, and On-Policy Distillation for Long-Horizon Post-Training

Limes Labs Research Workstream

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Abstract

Critic-free reinforcement learning, especially Group Relative Policy Optimization (GRPO), is attractive for reasoning-model post-training because it removes the learned value model and reduces memory pressure. This paper asks where that trade changes in long-horizon, variable-length, tool-using trajectories. Our working hypothesis is conditional: critic-based PPO is a better candidate when token- or state-level temporal credit is observable and covered, while GRPO remains compelling when group reward contrast is reliable and value modeling is weak or too expensive. OPD and OPSD are treated as complementary on-policy distillation objectives, not direct replacements for policy optimization. We formalize the method taxonomy, cost accounting, and failure modes, and run a CPU-only toy benchmark with known oracle advantages. Across a 20-seed, 18-case matrix, critic-style temporal-difference advantages have higher mean oracle-advantage correlation in 17 cases; a confidence-interval reading gives 16 clear critic-favorable cases, 1 near tie, and 1 clear group-favorable counterexample. The result is mechanism evidence, not a frontier-model leaderboard claim.

Keywords. post-training; reinforcement learning from verifiable rewards; PPO; GRPO; on-policy distillation; credit assignment; long-horizon agents.

1 Introduction

Long-horizon post-training for language-model agents is not merely long-context language modeling. A trajectory may include partial plans, tool calls, retrievals, edits, failed attempts, retries, compacted summaries, and final verifier feedback. In such traces, the question “did the rollout succeed?” is not the same as “which token or action helped?” A scalar terminal reward can be useful for ranking responses, but it can also broadcast the same update over helpful subgoals, harmful detours, and no-op padding.

This paper is a public Limes Labs workstream on the method question: *for variable-length, multi-step, tool-heavy trajectories, when does a critic/value-model setup recover useful temporal credit that a group-relative setup discards or blurs?* We deliberately avoid centering the work on one base model. Z.ai’s GLM-5.2 report is discussed only as an industry case study: it describes compacted, variable-length long-horizon rollouts and a move from group-wise optimization to critic-based PPO with token-level advantages [12]. That is not independent causal evidence that PPO beats GRPO; it is a motivating example of the problem structure studied here.

Contributions. We make four contributions. First, we separate PPO, GRPO, OPD, and OPSD by objective, signal granularity, and extra-model cost. Second, we give a cost and failure-mode checklist for fair post-training comparisons. Third, we introduce a small deterministic toy

environment with oracle advantages and compare group-relative and critic-style advantage estimators across a multi-seed matrix. Fourth, we state testable predictions for future neural and agentic experiments.

2 Definitions and Method Taxonomy

Let a policy produce a trajectory

$$\tau = (s_0, a_0, s_1, a_1, \dots, s_T), \quad (1)$$

where a state may include the prompt, partial response, tool outputs, verifier state, files, memory, or compacted context. Let $R(\tau)$ be an outcome reward, possibly combined with per-token KL, process rewards, or unit-test feedback.

2.1 PPO with a critic

PPO optimizes a clipped policy-gradient surrogate [9]:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t [\min(\rho_t(\theta)A_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)], \quad (2)$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}. \quad (3)$$

In actor-critic use, a value model V_ϕ can estimate one-step temporal difference errors

$$\delta_t = \nabla_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (4)$$

or generalized advantage estimates [8]:

$$A_t^{\text{GAE}(\gamma, \lambda)} = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \delta_{t+\ell}. \quad (5)$$

The critic is valuable only if the value target is learnable, sufficiently covered, and cheaper than the credit it recovers.

2.2 GRPO

GRPO samples a group of completions for the same prompt and normalizes rewards inside the group [2, 10]. A simplified response-level advantage is

$$\hat{A}_i^{\text{GRPO}} = \frac{R_i - \mu_G}{\sigma_G + \epsilon}, \quad (6)$$

where μ_G and σ_G are group statistics. This critic-free design is memory-efficient and effective in many verifiable-reward settings. Its long-horizon risk is intra-rollout information loss: one scalar can be broadcast across tokens with very different causal roles.

Table 1: Method taxonomy for long-horizon post-training.

Method	Objective	Signal granularity	granu-Extra signal	Main risk
PPO + critic	Reward/KL optimization	token/state	value model, reward, reference	critic cost and miscalibration
GRPO	Group-normalized RL	group/response scalar	reward or verifier, no critic	weak intra-rollout credit
OPD	Teacher distillation	token logits	teacher model	teacher mismatch and teacher cost
OPSD	Privileged distillation	self-token logits	privileged traces	privileged leakage and truncation

2.3 OPD and OPSD

OPD trains a student on student-sampled trajectories while a teacher provides dense supervision on those visited states [1, 3]. A typical objective is

$$L^{\text{OPD}} = \mathbb{E}_{x \sim D, y \sim \pi_s(\cdot | x)} \left[\sum_t D_{\text{KL}}(\pi_T(\cdot | x, y_{<t}) \| \pi_s(\cdot | x, y_{<t})) \right]. \quad (7)$$

OPSD removes the separate teacher model by making one model play a privileged teacher role and an ordinary student role over student rollouts [13]. These are dense distillation objectives. They can be excellent post-training tools, but they are not the same objective as reward optimization.

3 Cost Accounting

Fair comparisons should charge all hidden inputs, not only final benchmark score. For PPO, the bill includes value-model parameters, value targets, critic forward passes, optimizer state, old-policy log probabilities, KL control, and rollout storage. For GRPO, the bill includes group size, discarded completions, low-dispersion groups, and verifier calls. For OPD/OPSD, it includes teacher or privileged-context compute, dense logits, trajectory length inflation, and truncation handling. For agentic coding RL, it also includes anti-hacking filters, online tool-call monitors, LLM judges, sandboxing, blocked-call handling, and dummy observations when invalid calls are intercepted.

The near-term reporting unit for this workstream is:

$$\frac{\Delta \text{reward}}{\text{generated tokens} \times \text{wall clock} \times \text{extra model memory}}, \quad (8)$$

reported alongside reproducible artifacts and negative results.

4 Toy Experiment

4.1 Environment

The toy environment creates variable-length trajectories with three action types: **help**, **harm**, and **wait**. **help** increases a hidden progress score, **harm** decreases it, and **wait** consumes a token without changing progress. A terminal verifier returns success when the final score reaches a prompt-specific threshold. Because the finite dynamics are known, we can compute an oracle state-action advantage:

$$A^*(s_t, a_t) = -c + V^*(s_{t+1}, h - 1) - V^*(s_t, h), \quad (9)$$

Table 2: Selected rows from the 20-seed deep matrix. Delta is critic r minus group r ; CI is a 95% across-seed interval for delta.

Case	Group r	Critic r	Delta	CI	Reading
horizon_04_baseline	0.495	0.977	0.482	± 0.011	critic clear
horizon_16_long_wait	0.293	0.865	0.571	± 0.013	critic clear
critic_budget_002_full	0.352	0.367	0.015	± 0.027	near tie
critic_budget_128_full	0.352	0.906	0.554	± 0.017	critic clear
observability_blind	0.353	0.482	0.129	± 0.017	critic clear
blind_undercovered_counterexample	0.510	0.455	-0.055	± 0.015	group clear
sparse_hard_group08	0.310	0.910	0.600	± 0.008	critic clear

where c is token cost and h is remaining horizon.

4.2 Estimators and metrics

The group-relative estimator normalizes terminal returns inside each prompt group and broadcasts the resulting scalar to all tokens in the trajectory. The critic-style estimator fits a tabular value model from sampled returns-to-go and computes one-step temporal-difference advantages. We measure Pearson correlation with the oracle advantage, calibrated MSE after affine rescaling, raw MSE, sign accuracy, wait-token leakage, within-trajectory variance, zero-variance group fraction, and exact critic-state hit rate.

This is not a closed-loop PPO-vs-GRPO training benchmark. It is an estimator fidelity experiment under controlled dynamics.

5 Results

The canonical matrix uses 20 seeds and 18 fixed cases across horizon length, group size, critic budget, observability, sparse-reward regimes, and a designed counterexample. Critic-style TD has the higher mean correlation in 17 of 18 cases. A confidence-interval reading is more cautious: 16 cases are clearly critic-favorable, 1 is a near tie, and 1 is clearly group-favorable. The mean critic-minus-group correlation is 0.420. Table 2 reports representative cases; the full CSV and JSON are committed in `results/`.

Interpretation. The evidence supports the PPO-favorable hypothesis in this controlled toy: when progress state is observable and critic coverage is adequate, critic-style advantages track token-level oracle credit much better than response-level group scalars. The result also protects against dogma. In the coverage-limited case, the confidence interval crosses zero; in the blind and undercovered case, group-relative terminal outcome information is better than a poor value table.

6 Failure Modes and Testable Predictions

PPO/critic failures. Critics can overfit reward artifacts, become stale after policy drift, collapse on long sparse trajectories, or assign confident but wrong temporal credit.

Critic minus group correlation by case

Positive values favor critic-style TD; negative values favor group-relative advantage.

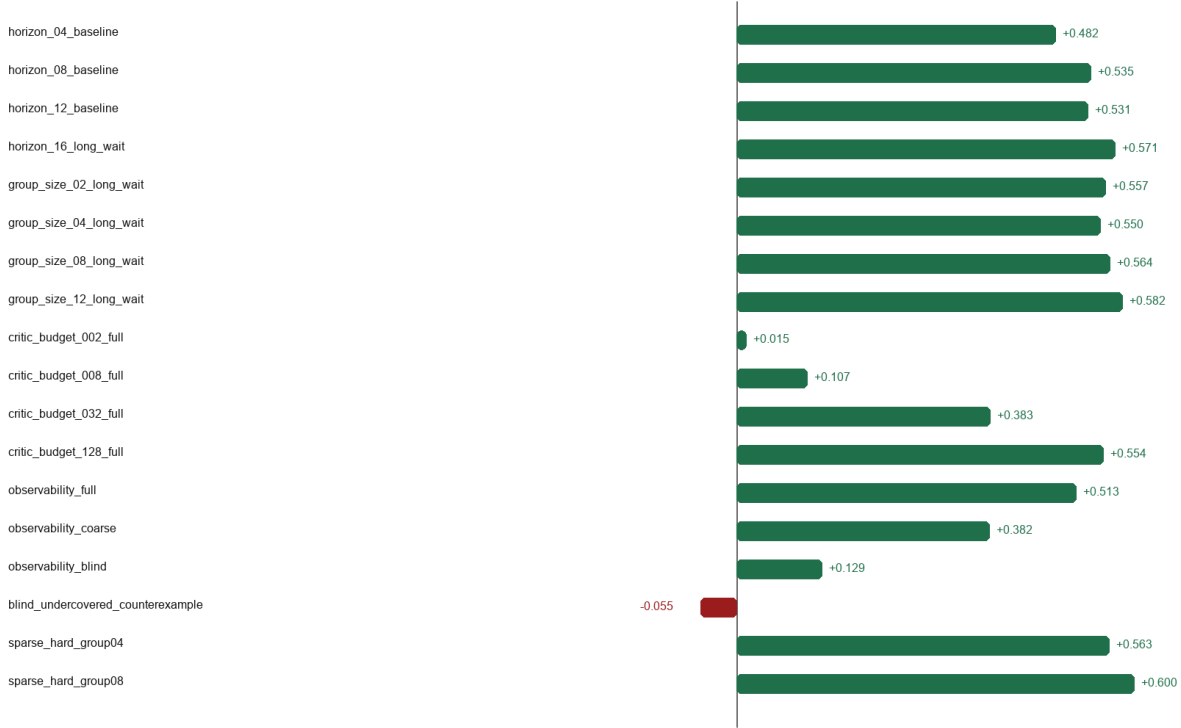


Figure 1: Mean critic-minus-group oracle-advantage correlation by case. Positive values favor critic-style temporal-difference advantages.

GRPO failures. GRPO can suffer zero-variance groups, low-dispersion amplification, response length bias, scalar reward broadcast over harmful tokens, and group composition confounding prompt difficulty with policy quality [4, 6, 7, 11, 14].

OPD/OPSD failures. Teacher/student thinking patterns may be incompatible; the teacher may offer no new information; on-policy rollouts may inflate in length and truncate; and privileged self-teacher traces may leak solution structure [3, 5, 13].

Predictions. Increasing horizon length and no-op padding should reduce the alignment between response-level group advantages and process labels. Increasing group size should help GRPO only when it increases meaningful reward dispersion. Adding process rewards should narrow the PPO/GRPO credit gap. Degrading critic observability should produce regimes where GRPO wins. OPD/OPSD should be most token-efficient when teacher signals are novel and rollout lengths are controlled.

7 Reproducibility

The core experiment code has no external Python dependencies. The canonical checks are:

```
python3 -m unittest discover -s tests
```

Critic coverage vs estimator advantage

A group-relative counterexample appears when critic state coverage is weak.

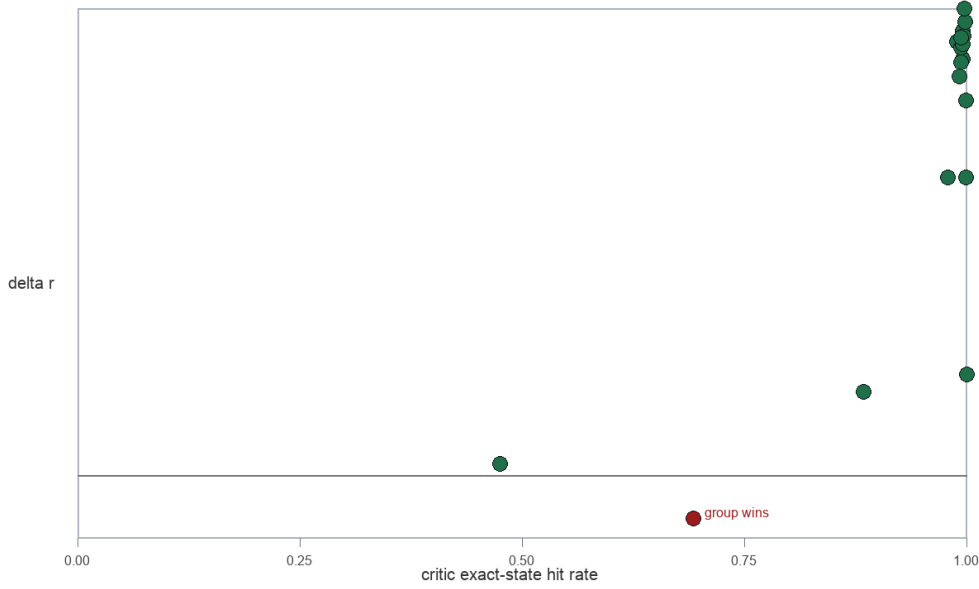


Figure 2: Critic state coverage versus estimator advantage. The group-relative counterexample occurs when the critic is blind and undercovered.

```
./scripts/run_smoke.sh
python3 -m experiments.deep_matrix \
  --output-json results/deep_matrix_20seed.json \
  --output-csv results/deep_matrix_20seed.csv \
  --output-md results/deep_matrix_20seed.md \
  --figures-dir results/figures
./scripts/build_latex_paper.sh
```

The LaTeX paper is built with Tectonic when available and emitted as the full paper PDF artifact under `public/`.

8 Limitations

The toy environment has a finite known state and a tabular critic. Real post-training adds neural approximation, KL controllers, reward-model errors, tool failures, distributed rollout systems, stale policies, optimizer details, and data contamination risks. The GRPO baseline is intentionally simple; the next benchmark should include leave-one-out baselines, length-normalized variants, DAPO/Dr. GRPO-style corrections, process rewards, and dynamic sampling before making stronger claims.

9 Conclusion

For long heterogeneous trajectories, critic-based PPO is a strong candidate when value targets are learnable and adequately covered. GRPO remains attractive when group reward contrast is reliable and critic modeling is too costly or poorly observed. OPD and OPSD are complementary dense-supervision tools that can transfer behaviors on student-visited states, but they should not be conflated with reward optimization. The right public research program is therefore not “PPO always beats GRPO,” but a set of falsifiable boundaries for when temporal credit is worth its cost.

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